Thalabus: Evidence-linked reasoning over parliamentary documents

- **Track:** Hack Apertus, Track 2B
- **Event format:** Online
- **Team:** `thalabus` — Luca Strozzi
- **Demo:** Two-minute recorded workflow; final recording link to be supplied with the submission
- **Report snapshot:** 2026-10-16

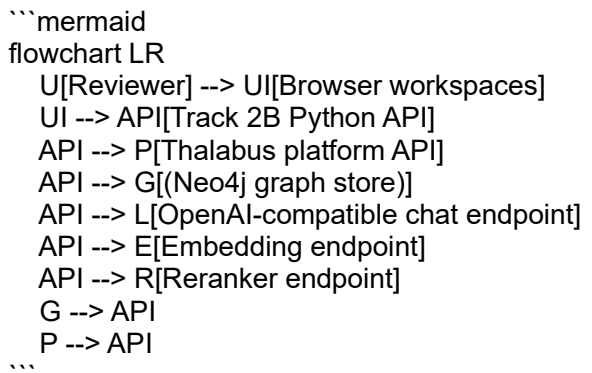
1. Summary

Parliamentary records are published as separate documents even when they concern the same affair. Finding an answer can require locating the original question, later committee or council material, and a response, while avoiding documents that merely share a person, institution, or topic. Thalabus Track 2B is a browser-based prototype that makes this path inspectable: documents are ingested into a graph-backed workflow, classified against a constrained parliamentary ontology, optionally expanded through graph evidence, and queried with citations to retrieved text.

The main contribution is not a claim of autonomous legal reasoning. The design premise is that this application and the Thalabus platform beneath it are built for local, on-premise installations in small data centres: no hyperscaler and no frontier LLM, with documents, graph and inference kept on infrastructure the operator administers. The application is an auditable workflow with visible proposal-versus-assignment state, document-level provenance, explicit rejection reasons, and a simple/advanced retrieval comparison. In the live development graph, re-categorising two Capaul-affair records moved `docid-411` from zero to 11 assigned categories and `docid-394` from 13 to 17. The nine-document case cluster now has eight categorised documents; `docid-413` remains the deliberate zero-category boundary. These are snapshot observations, not an accuracy benchmark. Advanced chat evaluation and a fresh knowledge-graph extraction of the cluster were still outstanding at this report snapshot.

2. Architecture

The same Python application can run in two configurations. Both the application and the platform are conceived for small, locally administered data centres, which is why the submitted target is **Plan Local** and why an open model that fits one machine is used instead of a frontier-scale service. In that design, Docker Compose runs the application, Neo4j, the Thalabus platform, embeddings, and
reranking services on infrastructure administered by the deploying organisation. The app connects to the platform through its API and keeps its own graph store separate from the platform database. Plan SaaS is a development and fallback profile: it runs the app locally and reaches a configured platform and model endpoints over the network. The live measurements in this report were taken with that development profile against the shared development graph, so they do not demonstrate the on-premise property; they are not a clean Plan Local installation.



The workspaces cover document ingestion, ontology inspection and category inference, assignment and extension approval, knowledge-graph extraction, graph exploration, and question answering. Category inference stores proposals on the document; only an explicit categorisation action creates `HAS_CATEGORY`/`OF_CATEGORY` assignments and promotes the
document. Proposed ontology definitions are marked `proposed` and excluded from inference until approved.

Chat simple mode embeds the question, searches document chunks, reranks candidates, applies a relevance threshold, and generates an answer from retained fragments with citations. Advanced mode adds category-affinity and shared-entity candidate discovery, checks entity bridges against both current document texts, and asks the chat model to judge each surviving document pair. Confirmed `RELATES` edges retain method, evidence, model, and seed provenance; rejected pairs are returned to the UI without an edge. Both modes preserve citations and surface degradation notes.

Target architecture and dependencies

The selected target is **on-premise**. Plan Local co-locates the application, graph database, platform services, and retrieval models on the organisation's own Docker host. It is designed to keep document processing and graph storage under that operator's control. This report does not claim sovereign-Swiss-cloud deployment or a verified air-gap: those require separate hosting and network evidence. Plan Local's release gate is still open because the two platform-image mirror digests and the platform owner's pull-and-run authorization are pending.

Plan Local's documented tested hardware class is Linux ARM64, NVIDIA DGX Spark/GB10, 128 GB
unified system memory, and at least 96 GB GPU memory. Its preflight enforces the hardware requirements. Other architectures are not validated. Docker/Compose, image availability, model weights, and the generated platform JWT keys are setup dependencies. At runtime the local plan uses the chat endpoint configured for the platform and in-network embedding and reranking services; the SaaS plan additionally requires network access, credentials, and an authorized Thalabus instance. No credentials are included in this report.

3. Use of Apertus

The Thalabus platform model configuration in the SaaS profile is `onprem-ai/Apertus-v1.5-70B-FP8`. The platform serves chat completions through its OpenAI-compatible endpoint. Track 2B app-side operations use the same API shape but read an independent model setting: the checked-in SaaS Compose default is `qwen-3.5-35b`, while its platform model card is configured for Apertus. Therefore the repo configuration alone does not prove that every Track 2B categorisation, extraction, pair-judging, or answer call ran on Apertus. The final demo/evaluation record must capture the resolved app model and show the Apertus route for any results claimed as Apertus-produced.

Prompts are application-owned; no fine-tuning, LoRA adapter, or quantisation step is performed by this project. The platform model is configured as FP8 by its model identifier. The local retrieval stack uses `Qwen/Qwen3-Embedding-0.6B` for embeddings and `BAAI/bge-reranker-v2-m3` for cross-encoder reranking; those are retrieval models, not substitutes for the Apertus chat model. Model weights and serving images are external build or deployment artefacts and are not vendored into the application source.

4. Data

The application repository tracks 39 hash-named sample PDFs in `data/sample/` (about 18.9 MB). The live development evaluation also used the Thalabus platform's `hackathon-temp` dossier,

which contained 40 document nodes at the measured snapshot. The report's case study is the Schaffhausen parliamentary inquiry *Kleine Anfrage 20 24 / 2 6*, represented by nine linked documents (`docid-389`, `391`, `394`, `402`, `405`, `410`, `411`, `413`, `414`). The files are source documents, not human-subject study data; they may contain names and other personal information appearing in public records. No human-subject recruitment or consent process was part of this work.

The repository does not currently contain a per-file source and redistribution-licence manifest for the PDFs. Their presence in the staging checkout is not evidence of permission to redistribute them. Before creating the public package, verify each source and licence, remove any material without a clear redistribution basis, and record the remaining provenance in `PROVENANCE.md`. The same release review must reconcile the Apache-2.0 template licence, the T2B open-source release attestation, and the still-unselected licence for the copied platform source slice. This report does not assert that these distinct components share one licence.

5. Evaluation

Evaluation was a mix of offline contract tests and a read-only inspection of the shared development graph. There was no held-out question set, labelled relevance set, answer-accuracy scoring, or controlled latency comparison. Consequently the results below establish exercised behaviour and observed state, not general retrieval quality.

Setup	Metric	Result
Simple RAG baseline	Quantitative answer quality / latency	Not benchmarked against a labelled test set
Advanced retrieval	Live end-to-end answer quality and expansion latency	Pending; the implementation and offline gate tests pass, but no post-change live comparison is claimed
Category inference and application	Live case-state change	`docid-411`: 0 → 11 categories; `docid-394`: 13 → 17; both status values became `Ready`
Document identity	Offline unit tests	6/6 passed, including platform-node reuse and refusal to adopt a foreign node
Categorisation	Offline unit tests	11/11 passed
Ontology extension	Offline unit tests	9/9 passed; proposed definitions stay out of inference until approval
Advanced expansion	Offline fake-store tests	0 failures; includes category ladder, co-mention gate, pair-judge verdicts, provenance, cache, and degradation paths

At the post-T1 snapshot, eight of the nine case documents had assigned categories. Counts were: `389` 11, `391` 10, `394` 17, `402` 10, `405` 11, `410` 10, `411` 11, `413` 0, and `414` 6. Across these records, entity counts ranged from 0 (`405`) to 60 (`410`). The `Parlament` ontology read on 2026-10-16 contained 49 definitions (27 approved and 22 proposed); proposed definitions had zero assignments, preserving the approval gate. These counts are from the shared graph and its dated ontology query, and must not be generalized to a clean deployment. Category assignment counts in this report are the later post-T1 document snapshot and are listed separately above.

The case exposed two concrete risks. First, `docid-413` remains uncategorised, so a category ladder cannot discover it. Second, the extracted entities on the rewritten `docid-394` text were stale at the measurement point. A fresh extraction of the eight-document target set and the live advanced-chat trace were still in progress; no claim that the full case trajectory was successfully reconstructed is made here.

6. Limitations

- The current case cluster has an over-merged affair entity: the same affair node is claimed by

documents that belong to different inquiries. A shared entity or category is a candidate signal, not proof of identity; the pair judge can also be wrong.
- Typed entity-to-entity extraction and cross-document event resolution are incomplete. The advanced mode uses candidate generation and an LLM pair check as a bounded substitute, not a validated legal-linking model.
- Text can change after extraction. Existing entity edges are not automatically proof that they describe the current text; the cluster must be re-extracted after a rewrite.
- Categories are ontology-constrained but may still be legitimate values applied to the wrong document. Proposed definitions are not classifications and require explicit approval.
- No statistical accuracy, calibration, demographic fairness, or multilingual quality study has been performed. No response-time result for live advanced mode is reported.
- Plan Local has not been established as a clean, reproducible release in this report; image digests, authorization, complete test deployment, and the release licensing/data audit remain gates. The shared development graph is mutable and contains unrelated corpora.

7. Reproducibility

The source baseline inspected for this report is commit `dd9dcf37b1fcd7c261a1968d292f17deba4b0fde` in the Track 2B checkout. The focused offline tests were run on Windows from that checkout with its Python environment; no random seed is used by those contract tests. Live graph numbers are a dated shared-database snapshot, not a fixture that a clean checkout recreates automatically.

For the documented local path, use Linux ARM64 on the tested DGX Spark hardware class, prepare
the model cache and JWT key pair, configure secrets in an untracked `.env`, then run:

```sh
make keys
make run
make health
```

To exercise the development fallback, use `make run PLAN=SaaS` with authorized endpoint credentials and a reachable Thalabus instance. Do not use that profile as evidence of an on-premise or air-gapped deployment. A clean release rehearsal must also pass `make lint`, `make verify`, the offline tests above, a fresh extraction and chat trace, and a source/data provenance review. The exact model IDs, image digests, hardware, data snapshot, and test output should accompany that final rehearsal.

8. Next steps

1. Complete the eight-document extraction and capture the live advanced-mode question, answer, citations, pair decisions, and elapsed time; compare against simple mode on a small labelled set of questions.
2. Improve affair identity resolution and prevent stale extraction links after document text changes; add regression cases for over-merged and uncategorised documents.
3. Finish the Plan Local image-digest, authorization, hardware, and clean-install gates; test with outbound traffic disabled if an air-gap claim is desired.
4. Complete per-document data provenance, component licensing, and public-package secret scans
before publishing the repository or its sample corpus.

Licence and release status

This is a technical report for the staging checkout, not a licence grant. The template carries

Apache-2.0, while copied T2B platform code and the sample documents have separate provenance
and licensing questions. The final public package must include a reviewed `PROVENANCE.md` and
the applicable licence decisions; until then, do not interpret this report or the staging
repository as permission to redistribute platform code, private data, or hosted-service access.

References

- [`README.md`](README.md) — product scope, run plans, operations, and packaging.
- [`docs/RUN_PLANS.md`](docs/RUN_PLANS.md) — deployment topology, hardware, and plan constraints.
- [`docs/CATEGORIZATION.md`](docs/CATEGORIZATION.md) — category proposal, assignment, and ontology-extension contracts.
- [`docs/KG_EXTRACTION.md`](docs/KG_EXTRACTION.md) — extraction pipeline and persistence model.
- [`docs/CHAT_QA.md`](docs/CHAT_QA.md) — simple and advanced retrieval, citations, and known evaluation gaps.
- [`../../../Documentation/Documents/active/events/2026_10_HACKATHON.T34.md`](../../../Documentation/Documents/active/events/2026_10_HACKATHON.T34.md) — live case snapshot and post-T1 caption figures.
- [`src/SLICE_PROVENANCE.md`](src/SLICE_PROVENANCE.md) — copied source scope and licensing prerequisites.